LESSON 01 / 10

A limit describes approach, not attendance

Write $\lim_{x\to a} f(x) = L$ when $f(x)$ stays arbitrarily close to L whenever x is sufficiently close to a, **with** $x \neq a$. The value $f(a)$ may be different, or may not exist at all.

For example,

$$\frac{x^2-1}{x-1} = x+1 \qquad (x \neq 1),$$

so its limit as $x \to 1$ is 2, even though the original expression is undefined at 1. Cancellation changes the formula only at the excluded point, which the limit already ignores.

The ε–δ definition makes "arbitrarily close" precise

The statement $\lim_{x\to a} f(x) = L$ means:

$$\forall \varepsilon > 0 \ \exists \delta > 0 : \quad 0 < |x-a| < \delta \implies |f(x)-L| < \varepsilon.$$

You choose an output tolerance ε; a working δ must keep every punctured input in that band inside the output tolerance. One successful δ can always be replaced by a smaller positive one.

WORKED EXAMPLE

A complete ε–δ proof for a linear function

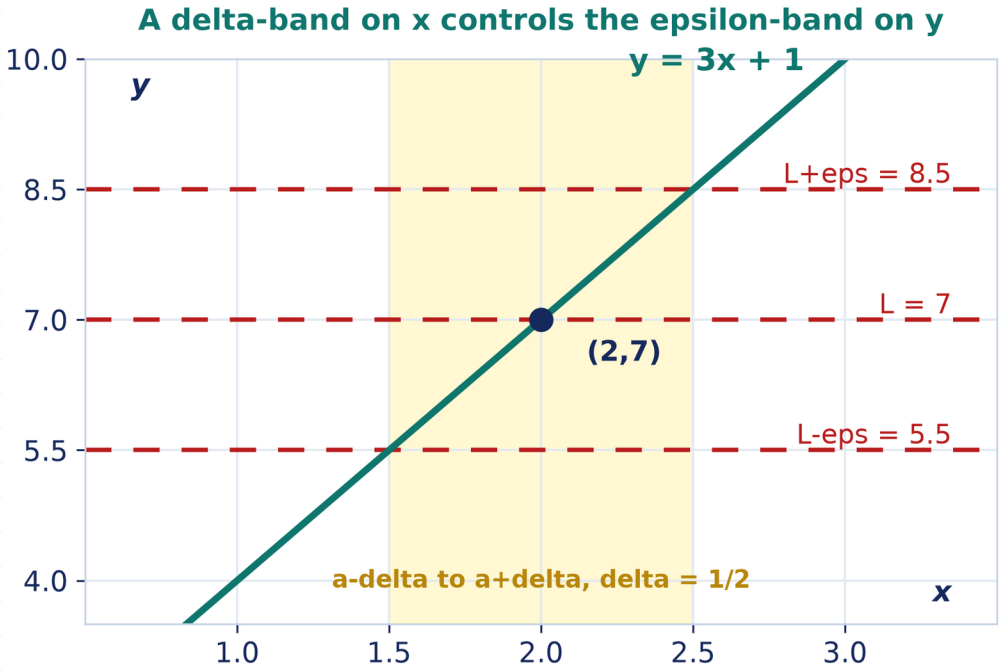
To prove $\lim_{x\to 2}(3x+1) = 7$, observe

$$|(3x+1)-7| = 3|x-2|.$$

Given $\varepsilon > 0$, choose $\delta = \varepsilon/3$. Then

$$0 < |x-2| < \delta \quad \implies \quad |(3x+1)-7| < 3\delta = \varepsilon.$$

For the illustrated choice $\varepsilon = 3/2$, the corresponding $\delta = 1/2$ works exactly.



READ THE FIGURE

For 1.5 < x < 2.5, the graph stays between 5.5 and 8.5. The puncture x != 2 matters in the definition, even though this line is defined there.

Two one-sided limits must agree

The two-sided limit exists and equals L exactly when

$$\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L.$$

A finite one-sided limit can exist on just one side when the domain lies only on that side, as with $\sqrt{x}$ at 0.

WORKED EXAMPLE

The greatest-integer function jumps at an integer

At $x = 2$,

$$\lim_{x\to 2^-} \lfloor x \rfloor = 1, \qquad \lim_{x\to 2^+} \lfloor x \rfloor = 2.$$

Therefore the two-sided limit does not exist. At a noninteger, the function is locally constant and its two-sided limit does exist.

COMMON TRAP

Defined, finite and convergent are different claims

A function can be defined at a and still have no limit there. It can have a limit without being defined there. And writing $f(x) \to \infty$ describes unbounded divergence, not a finite limit. State which claim you are making before calculating.

LESSON 02 / 10

Limit laws work after existence is known

If $f(x) \to L$ and $g(x) \to M$ with finite L, M, then sums, differences, products and constant multiples pass through the limit. If $M \neq 0$, then

$$\frac{f(x)}{g(x)} \to \frac{L}{M}.$$

For a product, the proof uses

$$|fg - LM| \le |f||g - M| + |M||f - L|.$$

A convergent f is locally bounded, so both terms can be made small. The quotient law requires the denominator's limit to be nonzero; it says nothing about a $0/0$ expression.

Factor out the cause of 0/0

Substitution gives $0/0$ only when both numerator and denominator vanish. Factor their common zero, cancel on a punctured neighbourhood, and then use the limit laws.

WORKED EXAMPLE

A cubic and quadratic sharing a zero

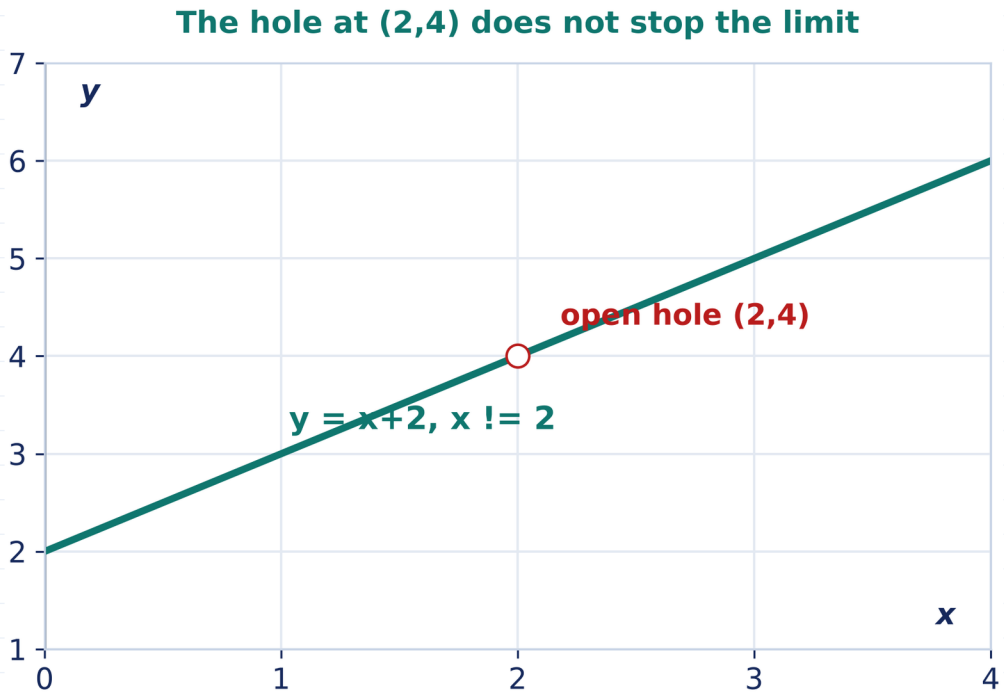
For $x \neq 1$,

$$\frac{x^3 - 1}{x^2 - 1} = \frac{(x-1)(x^2 + x + 1)}{(x-1)(x+1)} = \frac{x^2 + x + 1}{x + 1}.$$

Hence the limit as $x \to 1$ is $3/2$. More generally, for integer $n \ge 1$,

$$\frac{x^n - 1}{x - 1} = x^{n-1} + x^{n-2} + \cdots + x + 1 \longrightarrow n.$$

The right-hand sum has exactly n terms, each tending to 1.



READ THE FIGURE
The original quotient (x^2-4)/(x-2) is undefined at 2 but equals x+2 everywhere nearby. Both one-sided approaches point to height 4.

Conjugates turn radical 0/0 into ordinary algebra

WORKED EXAMPLE
Rationalise, cancel, then substitute
For $x \neq 0$,

$$\frac{\sqrt{1+x}-1}{x} = \frac{(1+x)-1}{x(\sqrt{1+x}+1)} = \frac{1}{\sqrt{1+x}+1} \longrightarrow \frac{1}{2}.$$

Similarly, as $x \to 4$,

$$\frac{\sqrt{x}-2}{x-4} = \frac{1}{\sqrt{x}+2} \longrightarrow \frac{1}{4}.$$

The conjugate removes the radical from the numerator without changing the expression on the permitted domain.

WORKED EXAMPLE
A direct calculation is enough when no form is indeterminate
For $\lim_{x\to 2}(5x-3)$, ordinary arithmetic gives 7. For $\lim_{x\to 0}(x^2-3x)/x$, cancel x on $x \neq 0$ to obtain $x-3 \to -3$. L'Hôpital's rule is unnecessary in both cases.

KEY RESULT
Preserve the punctured-domain statement
Write "for $x \neq a$" beside a cancellation. The simplified formula determines the limit, but it does not restore a value or change the original function's domain. This distinction becomes the definition of a removable discontinuity later.

LESSON 03 / 10

Two known bounds can force an unknown limit

If $g(x) \le f(x) \le h(x)$ near a, and g, h both tend to L, then $f(x) \to L$. For an output tolerance ε, eventually both outer functions lie in $(L-\varepsilon, L+\varepsilon)$, leaving no other place for f.

WORKED EXAMPLE

Bounded oscillation multiplied by a vanishing factor

For $x \neq 0$,

$$-x^2 \le x^2 \sin\tfrac{1}{x} \le x^2.$$

Both bounds tend to 0, so $x^2\sin(1/x) \to 0$. Likewise,

$$\left|x\sin\tfrac{1}{x}\right| \le |x| \longrightarrow 0.$$

The oscillating factor has no limit by itself, but its boundedness is enough for the squeeze.

The fundamental trigonometric limit comes from areas

For $0 < x < \pi/2$, comparing triangle and sector areas gives

$$\tfrac{1}{2}\sin x < \tfrac{1}{2}x < \tfrac{1}{2}\tan x.$$

Rearranging,

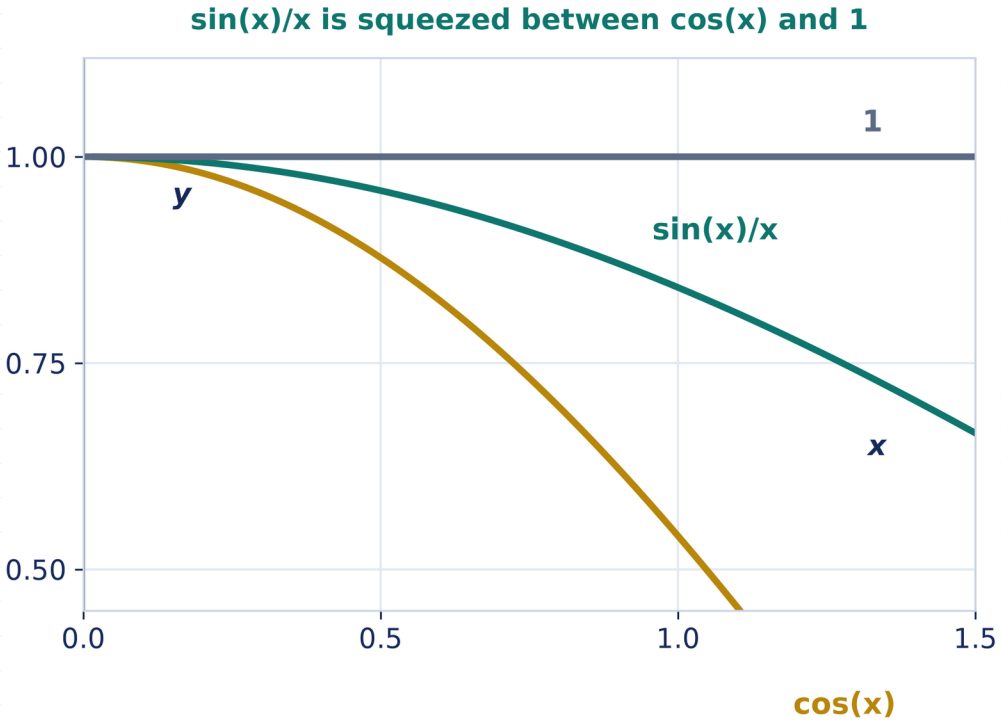
$$\cos x < \frac{\sin x}{x} < 1.$$

Since $\cos x \to 1$, the squeeze gives $\sin x/x \to 1$ from the right. The quotient is even, so the left limit is also 1.

COMMON TRAP

Radians and matching arguments are essential

The area proof and the standard trigonometric limits use radians. A formula $\sin u/u \to 1$ requires the **same** u above and below, with $u \to 0$ and $u \neq 0$. Do not replace $\sin 5x$ by $5x$ and then forget the factor 5.



READ THE FIGURE
On 0 < x < pi/2, the middle curve approaches 1 because both outer values approach 1. Evenness supplies the left-hand conclusion.

The same standard limit yields

$$\lim_{x\to 0}\frac{\tan x}{x}=1, \qquad \lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2}.$$

For the second, use $1-\cos x=2\sin^2(x/2)$ or multiply by $1+\cos x$. Consequently,

$$\frac{\cos x-1}{\sin^2 x}=-\frac{1}{1+\cos x}\longrightarrow -\frac{1}{2}.$$

WORKED EXAMPLE

Match the small argument with its denominator

$$\frac{\sin 5x}{3x}=\frac{5}{3}\cdot\frac{\sin 5x}{5x}\longrightarrow\frac{5}{3}.$$

Also $1-\cos 2x=2\sin^2 x$, so

$$\frac{1-\cos 2x}{x\sin x}=2\cdot\frac{\sin x}{x}\longrightarrow 2.$$

WORKED EXAMPLE

A third-power trigonometric limit without series

For $x\neq 0$,

$$\frac{\tan x-\sin x}{x^3}=\frac{1}{\cos x}\cdot\frac{\sin x}{x}\cdot\frac{1-\cos x}{x^2}\longrightarrow 1\cdot 1\cdot\frac{1}{2}=\frac{1}{2}.$$

Each factor has a separately justified finite limit.

LESSON 04 / 10

The two first-order standards

Near $x = 0$,

$$\lim_{x\to 0} \frac{e^x - 1}{x} = 1, \qquad \lim_{x\to 0} \frac{\ln(1+x)}{x} = 1.$$

They describe the first-order behaviour of the exponential and logarithm. Substitution rescales them: for fixed k,

$$\frac{e^{kx} - 1}{x} = k \cdot \frac{e^{kx} - 1}{kx} \longrightarrow k.$$

For $a > 0$, writing $a^x = e^{x \ln a}$ gives

$$\lim_{x\to 0} \frac{a^x - 1}{x} = \ln a.$$

Thus $(2^x - 1)/x \to \ln 2$ and $(e^{3x} - 1)/x \to 3$.

The number e is a continuous exponential limit

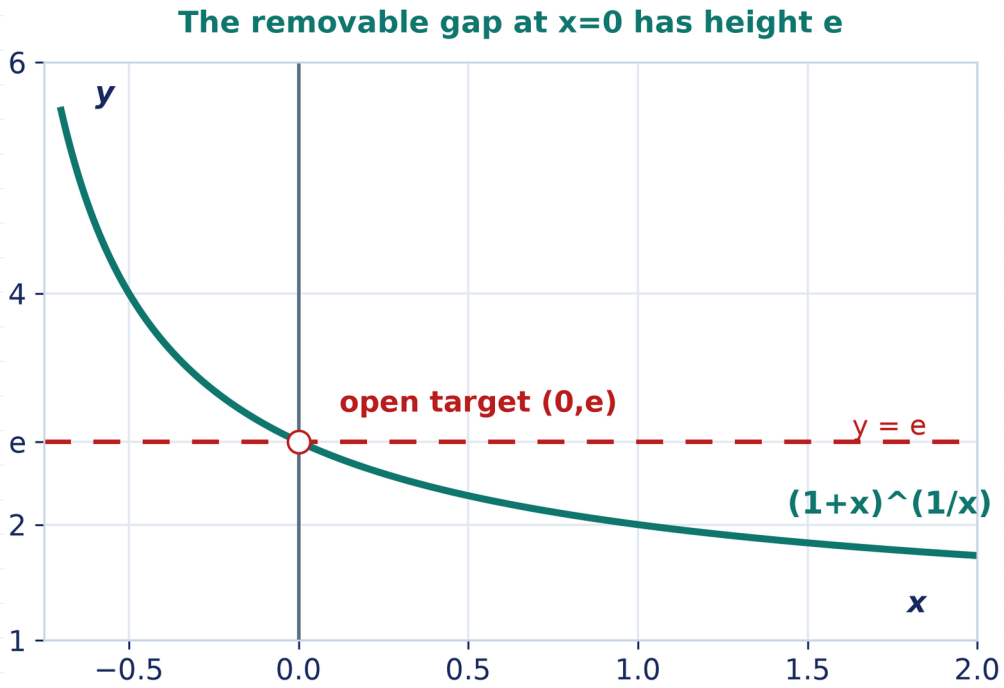
For $x > -1$ and $x \neq 0$, set

$$y = (1 + x)^{1/x}.$$

Taking logarithms and using continuity of $\ln$ and $\exp$,

$$\ln y = \frac{\ln(1+x)}{x} \longrightarrow 1 \quad \Longrightarrow \quad y \longrightarrow e.$$

Replacing x by $1/x$ also gives $(1 + 1/x)^x \to e$ as $x \to \infty$. More generally, $(1 + 3/x)^x \to e^3$. By the quotient law, $(1 + 1/n)^n / n \to 0$; this needs no Stolz–Cesàro argument.



READ THE FIGURE

The formula is undefined at 0, but the logarithm of the expression tends to 1 from both sides. Continuity of exp therefore forces the limit e.

Convert a 1^∞ form into a product before taking the limit

If $f(x) \to 0$, $1 + f(x) > 0$ near the approach, and $f(x)g(x) \to L$, then

$$(1 + f(x))^{g(x)} = \exp(g(x)\ln(1 + f(x))) \longrightarrow e^{L},$$

because, interpreting the last quotient by its removable value 1 if $f(x) = 0$,

$$g(x)\ln(1 + f(x)) = [f(x)g(x)]\frac{\ln(1 + f(x))}{f(x)} \longrightarrow L.$$

WORKED EXAMPLE

Extract the exponent coefficient exactly

As $x \to 0$,

$$(1 + 2x)^{3/x} \longrightarrow e^{6},$$

since $f(x) = 2x$, $g(x) = 3/x$, and $f(x)g(x) = 6$. The base tends to 1, but the exponent grows rapidly enough to preserve a finite non-1 limit.

COMMON TRAP

Do not substitute 1^∞ = 1

The forms 1^{∞}, 0^{0} and ∞^{0} are indeterminate because base and exponent compete. Take a logarithm first, and check that the logarithm's limit exists before exponentiating. Also record the domain condition $1 + f(x) > 0$ for a real variable exponent.

LESSON 05 / 10

At infinity, divide by the dominant power

For a rational function, equal highest degrees give the ratio of leading coefficients. Lower-degree terms disappear after division.

WORKED EXAMPLE

Normalize numerator and denominator by the same power

$$\frac{3x^2 - x + 4}{2x^2 + 5} = \frac{3 - 1/x + 4/x^2}{2 + 5/x^2} \longrightarrow \frac{3}{2} \qquad (x \to \infty).$$

Dividing by x instead of x^2 would leave a more complicated expression; choose the highest power present in the denominator after comparing degrees.

Growth hierarchies settle many apparent infinities

For fixed $p > 0$ and $a > 1$,

$$\ln x \ll x^p \ll a^x \qquad \text{as } x \to \infty.$$

In particular, $\ln x / x \to 0$ and $x / e^x \to 0$. These are not products of separate finite limits: they result from the relative growth of the numerator and denominator.

Limits define three kinds of asymptote

- **Horizontal:** $f(x) \to L$ as $x \to \infty$ or $x \to -\infty$ gives $y = L$ on that side.
- **Vertical:** a one-sided infinite approach at finite $x = a$ gives $x = a$.
- **Oblique:** $f(x) - (mx + b) \to 0$ gives $y = mx + b$, with $m \neq 0$.

WORKED EXAMPLE

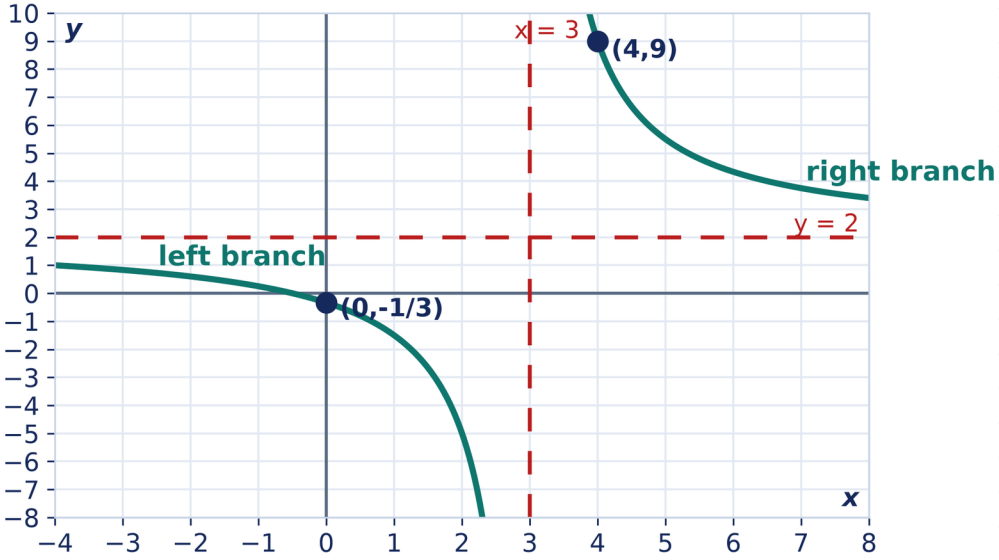
Read a hyperbola from two limits

For

$$f(x) = \frac{2x + 1}{x - 3} = 2 + \frac{7}{x - 3} \qquad (x \neq 3),$$

$f(x) \to 2$ as $x \to \pm\infty$, so $y = 2$ is horizontal. The residual term is unbounded near $x = 3$, so $x = 3$ is vertical.

The decomposition 2 + 7/(x-3) explains both asymptotes



READ THE FIGURE

Large x makes 7/(x-3) vanish, while x near 3 makes it unbounded. The two branches must be checked separately near the excluded point.

COMMON TRAP

A cancelled factor makes a hole, not a vertical asymptote

For

$$\frac{x+2}{x^2-x-6}=\frac{x+2}{(x-3)(x+2)}=\frac{1}{x-3}\qquad (x\neq 3,\ -2),$$

there is a removable hole at $x=-2$ and a vertical asymptote only at $x=3$. Always factor before declaring every denominator zero a pole.

LESSON 06 / 10

Differentiate only a genuine 0/0 or ∞/∞ quotient

Under the standard differentiability and nonzero-denominator-derivative hypotheses, if f/g has form $0/0$ or ∞/∞ and

$$\lim_{x\to a}\frac{f'(x)}{g'(x)}$$

exists finitely or as a signed infinity, then f/g has the same limit. The rule also applies to one-sided limits and limits at infinity. If the derivative ratio still has an indeterminate form, check the hypotheses again before applying the rule another time.

WORKED EXAMPLE

Two applications, then a standard limit

For

$$\lim_{x\to 0}\frac{x-\sin x}{x^3},$$

the form is $0/0$. Differentiating numerator and denominator gives

$$\frac{1-\cos x}{3x^2},$$

still $0/0$. A second application gives

$$\frac{\sin x}{6x}\longrightarrow\frac{1}{6}.$$

The repeated differentiation is on numerator and denominator separately; this is not the quotient rule.

WORKED EXAMPLE

A negative third-power tangent limit

For $(x-\tan x)/x^3$, two L'Hôpital steps lead to

$$\frac{-2\sec^2 x\tan x}{6x}=-\frac{\sec^2 x}{3}\cdot\frac{\tan x}{x}\longrightarrow-\frac{1}{3}.$$

The last step uses a standard limit rather than another unnecessary differentiation.

Other indeterminate forms must first become quotients

- For $0\cdot\infty$, write $fg=f/(1/g)$ or $g/(1/f)$.
- For $\infty-\infty$, combine fractions or rationalize.
- For 0^0, ∞^0 and 1^∞, take logarithms and evaluate the resulting product.

WORKED EXAMPLE

A $0\cdot\infty$ form and an $\infty-\infty$ form

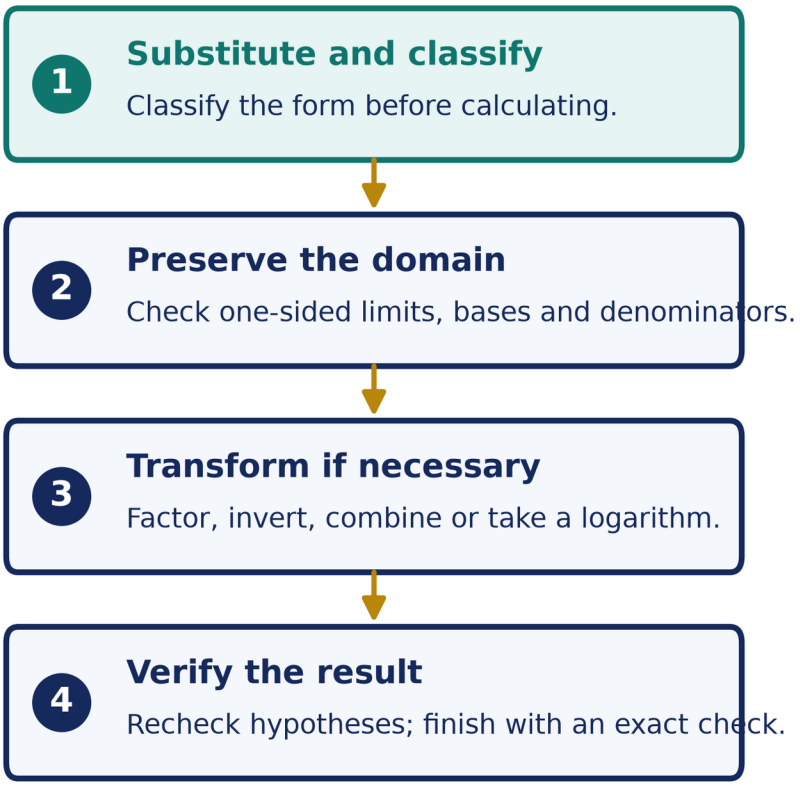
As $x\to 0^{+}$,

$$x\ln x=\frac{\ln x}{1/x}\quad\Longrightarrow\quad\frac{1/x}{-1/x^{2}}=-x\longrightarrow 0.$$

As $x\to\infty$,

$$\sqrt{x^{2}+x}-x=\frac{x}{\sqrt{x^{2}+x}+x}=\frac{1}{\sqrt{1+1/x}+1}\longrightarrow\frac{1}{2}.$$

Transform the form before choosing the rule



READ THE FIGURE

L'Hopital is one tool inside a domain-aware strategy, not a substitute for classifying the expression.

COMMON TRAP

Not every zero in the numerator is 0/0

For $\sin x/(x^{2}+1)$ as $x\to 0$, the form is $0/1$, so the limit is simply 0. For $1/x$ as $x\to 0$, the form is $1/0$, not an invitation to differentiate; the two-sided behaviour is unbounded with opposite signs.

LESSON 07 / 10

Three conditions must all hold

A function f is continuous at a when

$$f(a) \text{ is defined}, \qquad \lim_{x\to a} f(x) \text{ exists}, \qquad \lim_{x\to a} f(x) = f(a).$$

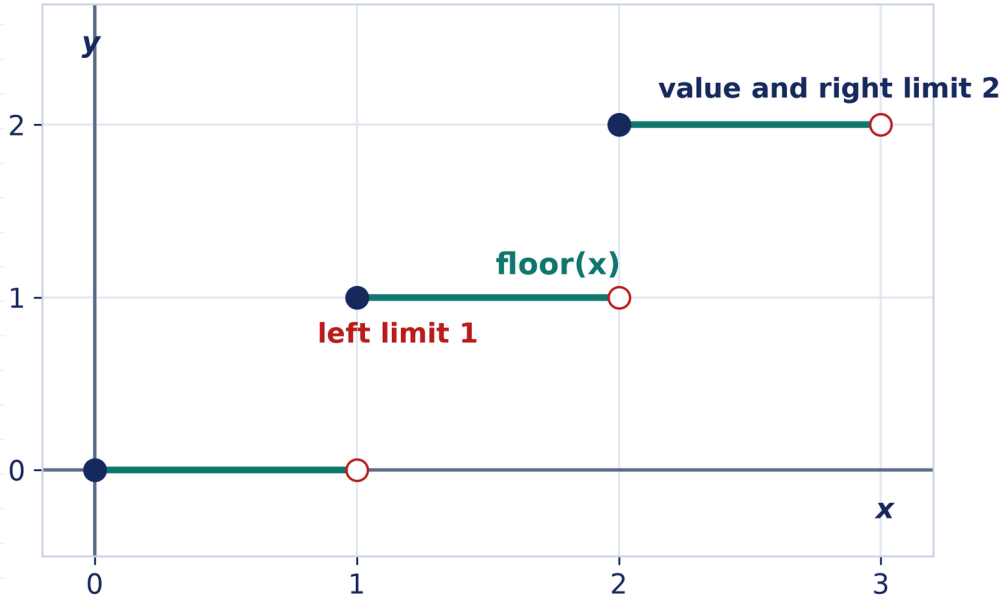
On an interval, continuity means this condition at every point, using one-sided limits at included endpoints. On a domain with isolated points or excluded points, always interpret the statement relative to that domain.

A limit can exist without continuity: if $f(x) = x^2$ for $x \neq 1$ but $f(1) = 3$, then the limit at 1 is 1 while the value is 3. The discontinuity is removable by redefining $f(1) = 1$.

Name the failure precisely

- **Removable:** a finite two-sided limit exists, but the value is missing or different.
- **Jump:** both one-sided limits are finite but unequal.
- **Infinite:** at least one side is unbounded.
- **Oscillatory:** nearby values fail to settle, as for $\sin(1/x)$ at 0.

A greatest-integer step has unequal one-sided limits



READ THE FIGURE
At x=2 the left limit is 1, while the value and right limit are 2. This is a jump, not a removable hole.

Algebra and composition preserve continuity on their domains

Sums, differences, products and constant multiples of continuous functions remain continuous. A quotient is continuous where its denominator is nonzero, and a composition is continuous when the inner output belongs to the outer function's domain. Consequently polynomials, exponentials, sine and cosine are continuous everywhere; logarithms and roots are continuous on their domains.

The function $|x|$ is continuous at 0 because both sides approach $|0| = 0$. Its corner concerns differentiability, which is a separate condition studied in the next chapter.

WORKED EXAMPLE

Match the value and the two one-sided limits

For $x \le 2$, let $f(x) = kx + 1$; for $x > 2$, let $f(x) = 3x - 1$. At 2,

$$f(2) = 2k + 1, \qquad \lim_{x \to 2^-} f(x) = 2k + 1, \qquad \lim_{x \to 2^+} f(x) = 5.$$

Continuity therefore requires $2k + 1 = 5$, so $k = 2$. With this value, all three quantities equal 5.

COMMON TRAP

A formula being defined is not enough

The floor function has a value at every integer but is discontinuous there. Conversely, $1/x$ is continuous at every point of its domain $(-\infty, 0) \cup (0, \infty)$, even though it has an infinite discontinuity at the excluded point 0. Always distinguish "continuous on its domain" from "continuous everywhere on $\mathbb{R}$."

LESSON 08 / 10

Continuity prevents an interval from being skipped

If f is continuous on $[a,b]$ and N lies between $f(a)$ and $f(b)$, then some $c\in[a,b]$ satisfies $f(c)=N$. In particular, **Bolzano's root theorem** says

$$f(a)f(b)<0 \quad\Longrightarrow\quad f(c)=0 \text{ for some } c\in(a,b).$$

The proof can be organized by bisection: repeatedly keep a half-interval whose endpoints straddle the target. Nested intervals locate a point where continuity forces the target value.

WORKED EXAMPLE

A sign change proves a root, and monotonicity makes it unique

For $f(x)=x^3+x-1$,

$$f(0)=-1<0, \qquad f(1)=1>0.$$

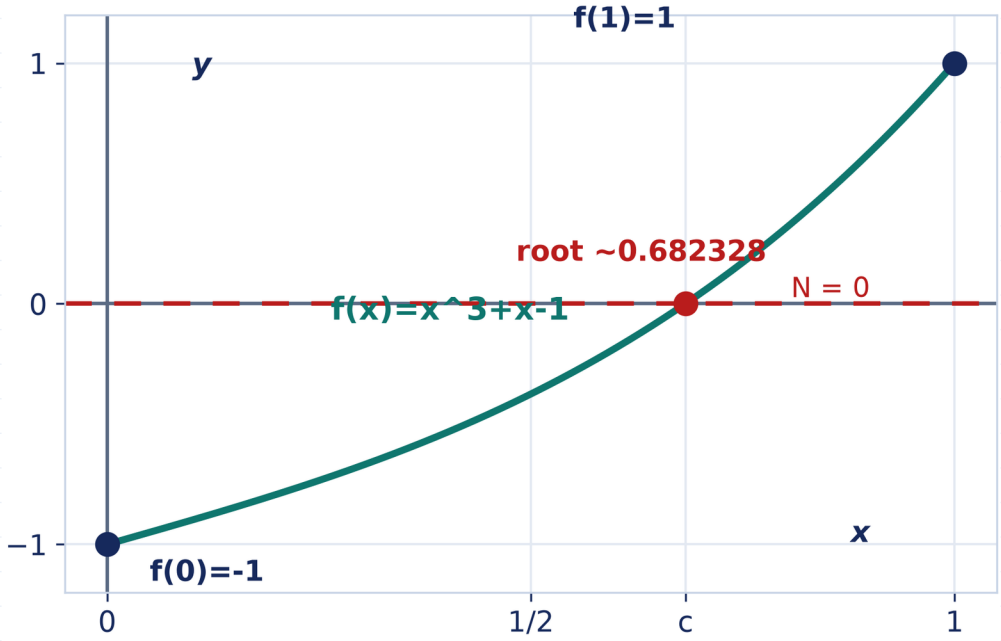
Thus a root lies in $(0,1)$. Moreover, x^3 and x are strictly increasing on $[0,1]$, so their sum is strictly increasing and the root there is unique. Numerically it is approximately 0.6823278038.

For a first bisection step,

$$f\left(\tfrac{1}{2}\right)=-\tfrac{3}{8}<0, \qquad f\left(\tfrac{3}{4}\right)=\tfrac{11}{64}>0,$$

so the root lies in $(1/2, 3/4)$.

A continuous curve cannot avoid the zero level



READ THE FIGURE

Opposite endpoint signs force at least one crossing. On this interval strict increase also rules out a second crossing.

Build an auxiliary function for existence questions

The equation $\cos x = x$ becomes $g(x) = \cos x - x = 0$. Since

$$g(0) = 1 > 0, \qquad g\left(\tfrac{\pi}{2}\right) = -\tfrac{\pi}{2} < 0,$$

a solution lies in $(0, \pi/2)$; its approximate value is 0.7390851332.

The same construction proves a fixed-point theorem. If $f : [0,1] \to [0,1]$ is continuous, put $g(x) = f(x) - x$. Then

$$g(0) = f(0) \geq 0, \qquad g(1) = f(1) - 1 \leq 0.$$

If either endpoint is zero, that endpoint is fixed; otherwise IVT gives an interior zero of g. Hence some c satisfies $f(c) = c$.

COMMON TRAP

IVT gives existence under continuity, not a converse

A sign change does not say how many roots exist unless monotonicity or another argument is added. Conversely, equal endpoint signs do not prove there is no root: x^2 has equal positive values at -1 and 1 but has a root at 0. IVT can also fail without continuity.

LESSON 09 / 10

Split at every definition change

Absolute values, greatest-integer functions, fractional parts and piecewise formulas require one-sided analysis at their breakpoints. For example,

$$\lim_{x\to 0^-}\frac{|x|}{x}=-1, \qquad \lim_{x\to 0^+}\frac{|x|}{x}=1,$$

so the two-sided limit does not exist. The fractional part $\{x\}=x-\lfloor x\rfloor$ has left limit 1 and right limit 0 at every integer.

WORKED EXAMPLE

Two standard limits and a value determine two parameters

Suppose a function is defined by

$$\frac{\sin(ax)}{x}\quad (x<0), \qquad b\quad (x=0), \qquad \frac{e^{2x}-1}{x}\quad (x>0).$$

Its left limit is a, its right limit is 2, and its value is b. Continuity at 0 requires all three to agree, hence

$$a=2, \qquad b=2.$$

Checking only $a=2$ would make the two-sided limit exist but would not ensure continuity unless $b=2$ as well.

Choose the smallest sufficient method

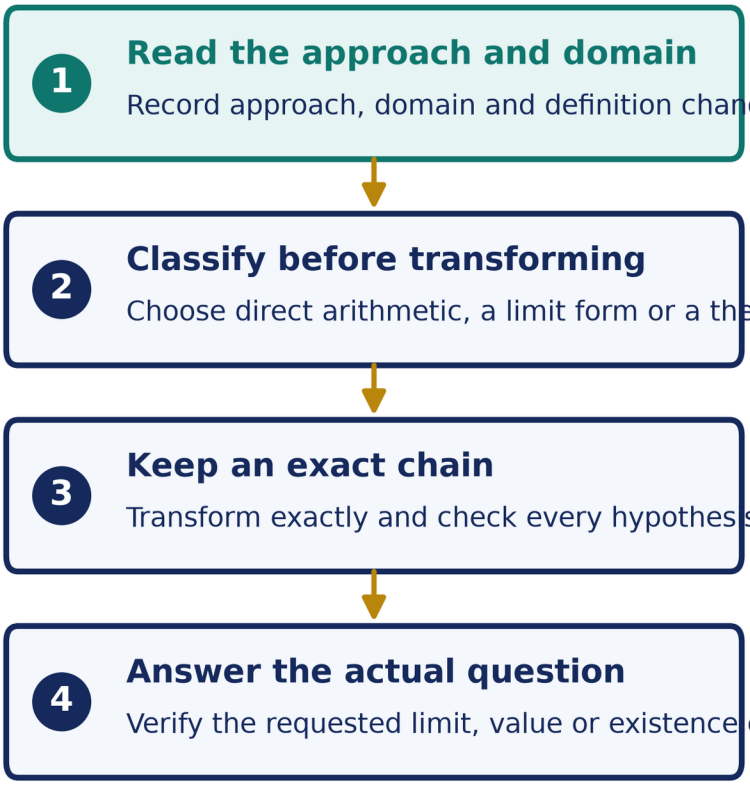
Match the expression to a reliable limit tool

Observation	Method	Essential check
Finite substitution	Direct limit laws	denominator limit nonzero
Polynomial or radical 0/0	Factor or conjugate	cancel only off the target
Bounded oscillation	Squeeze	both bounds share a limit
sin, exp or log standard	Match small argument	same argument and coefficient
0/0 or infinity/infinity	L'Hopital	hypotheses after every step
1^infinity or other power	Take logarithms	positive base; exponentiate last
Piecewise continuity	Match left, right and value	all three quantities agree
Root or intermediate value	Build an auxiliary function	continuity on the whole interval

READ THE FIGURE

The form suggests a method, but the hypotheses decide whether that method is valid.

A repeatable limit-and-continuity workflow



READ THE FIGURE

Most errors come from answering a different question from the one asked, not from the final arithmetic.

KEY RESULT

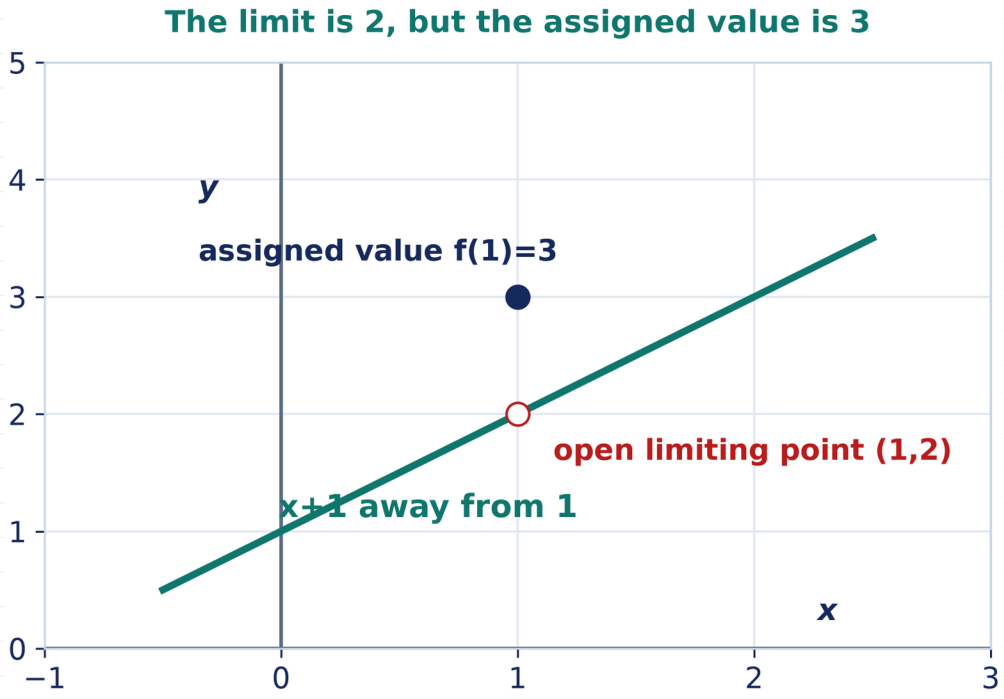
Final limit-and-continuity self-check

Did I preserve excluded points and one-sided restrictions? Did every cancellation occur only off the target? Did I normalize the correct small argument or highest power? Did every L'Hôpital step remain a permitted form? For continuity, do left limit, right limit and value all agree? For IVT, is the function continuous on the entire closed interval, and am I claiming existence rather than unsupported uniqueness?

LESSON 10 / 10

A hole can have a limit without having the right value

Consider $(x^2-1)/(x-1)$ for $x \neq 1$, and separately assign the value $f(1)=3$. Away from 1, the quotient equals $x+1$, so both approaches point to 2. The assigned value 3 sits above the limiting height.



READ THE FIGURE

The open point controls the limit; the closed point controls continuity. Redefining the closed point at (1,2) would remove the discontinuity.

Visual check: at $x=1.001$, the original quotient is 2.001 exactly. The approach is toward 2, regardless of the separately assigned value.

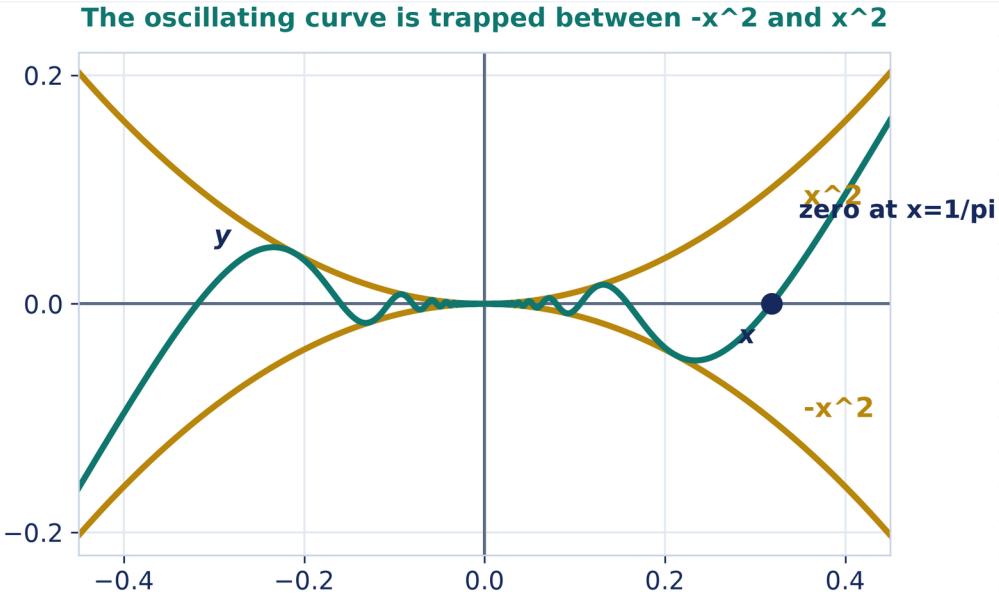
KEY RESULT

Second read: three different objects

The original domain excludes 1, the two-sided limit is 2, and the declared value is 3. Keeping those three claims separate prevents both algebra and continuity errors.

Squeeze makes oscillation irrelevant

For $x \neq 0$, the graph of $x^2\sin(1/x)$ oscillates increasingly often near 0. Nevertheless its absolute value never exceeds x^2, so it is trapped between the two smooth parabolas $-x^2$ and x^2.



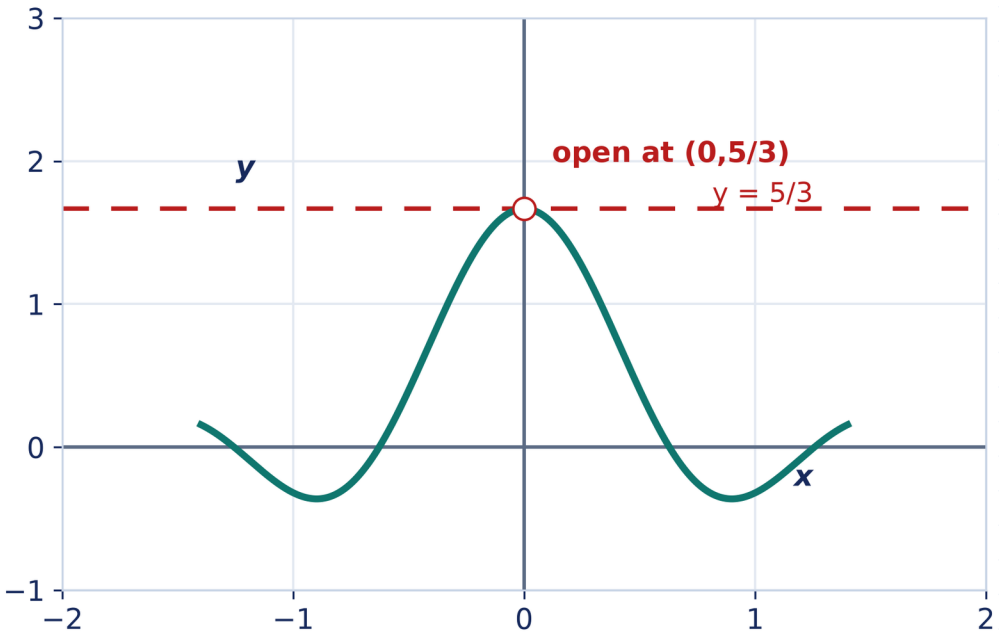
READ THE FIGURE
Frequency does not defeat the squeeze: both smooth bounds have height 0 at the origin, so the middle graph is forced to the same limit.

Visual check: at $x = 2/(5\pi)$, $\sin(1/x) = 1$, so the middle curve touches x^2. At $x = 1/\pi$, it touches zero. Both observations agree with $-x^2 \le f(x) \le x^2$.

A standard trigonometric limit has a visible target

The quotient $\sin(5x)/(3x)$ is undefined at 0. Away from 0, it equals $(5/3)\sin(5x)/(5x)$, so the limiting height is $5/3$.

The removable gap has height 5/3



READ THE FIGURE
Oscillations away from zero do not change the local target. Matching 5x in the sine and denominator produces the factor 5/3 exactly.

Visual check: at $x = 0.1$, the value is $\sin(0.5)/0.3 \approx 1.598085$, already near 1.666667. The graph and the standard-limit calculation identify the same hole.

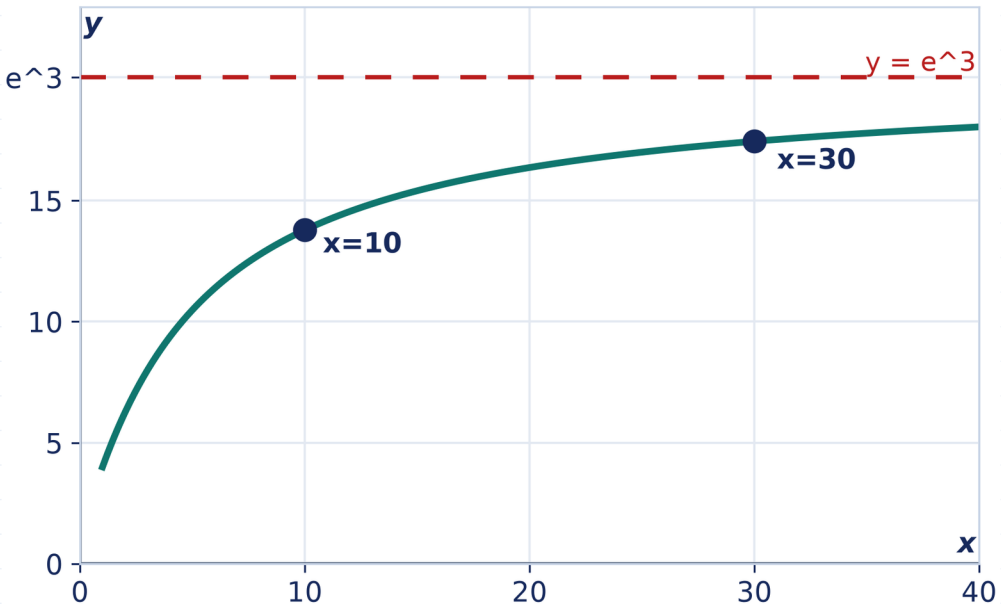
A base approaching 1 can still have a nontrivial limit

For $x > 0$, the base $1 + 3/x$ tends to 1 while the exponent x tends to infinity. The logarithm resolves the competition:

$$\ln\left(1+\tfrac{3}{x}\right)^{x} = 3\,\frac{\ln(1+3/x)}{3/x} \longrightarrow 3.$$

Therefore $(1+3/x)^{x} \to e^{3}$.

The growing exponent pulls the expression toward e^3



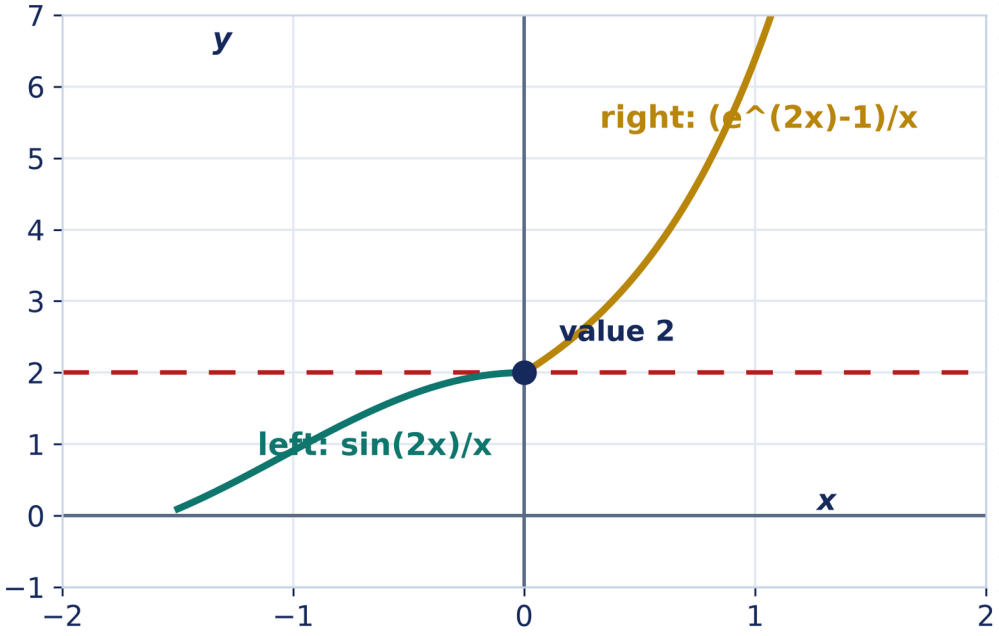
READ THE FIGURE
The horizontal asymptote is e^3, not 1. The product of the small base increment 3/x and the large exponent x remains 3.

Visual check: $1.3^{10} \approx 13.78585$ and $1.1^{30} \approx 17.44940$ lie below $e^{3} \approx 20.08554$, consistent with the graph's approach from below.

Continuity needs the left, right and value to meet

Use the three-piece formula from the mixed-strategy lesson with $a = b = 2$. The left formula is $\sin(2x)/x$, the right formula is $(e^{2x} - 1)/x$, and the value at 0 is 2.

Two different formulas can join continuously



READ THE FIGURE
The left limit, right limit and assigned value all equal 2. The formulas need not be the same on the two sides for continuity to hold.

Visual check: the left standard limit is 2, the right standard limit is 2, and the plotted point is exactly at their common height. If b were changed, only that point would move and continuity would fail.

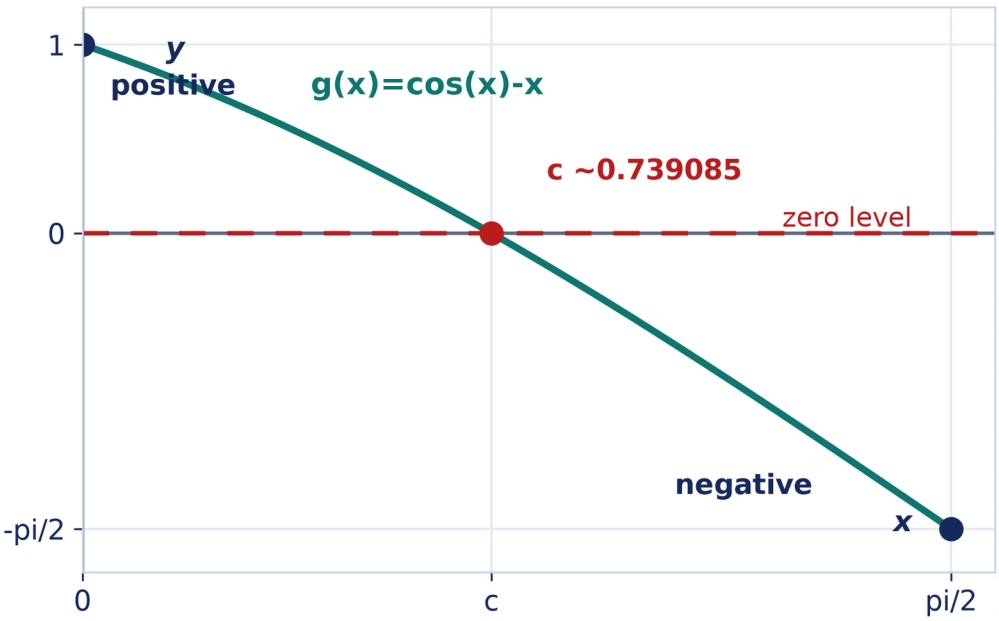
IVT turns opposite endpoint heights into a crossing

For $g(x) = \cos x - x$ on $[0, \pi/2]$,

$$g(0) = 1, \qquad g\left(\tfrac{\pi}{2}\right) = -\tfrac{\pi}{2}.$$

The continuous graph therefore crosses zero at least once. Its crossing is the Dottie number, approximately 0.7390851332.

The continuous difference cos(x)-x must cross zero



READ THE FIGURE
Continuity supplies the crossing. The IVT statement alone does not supply its decimal value or prove uniqueness.

KEY RESULT

Read every graph with a mathematical question

An open point asks for a limit; a closed point asks for continuity. Outer curves ask for a squeeze; an asymptote asks about growth; a sign change asks for IVT. The picture organizes the calculation, while exact substitution and the theorem hypotheses certify it.

LESSON 01 / 05

Heine's criterion turns function limits into sequence limits

For a function defined near a,

$$\lim_{x\to a} f(x) = L$$

holds exactly when **every** sequence $x_n \to a$ with $x_n \neq a$ satisfies $f(x_n) \to L$. To prove nonexistence, two sequences with different output limits are enough. To prove existence, checking only a favourite sequence is not enough; the statement must cover every approach.

WORKED EXAMPLE

Two sequences disprove the limit of sin(1/x)

Take

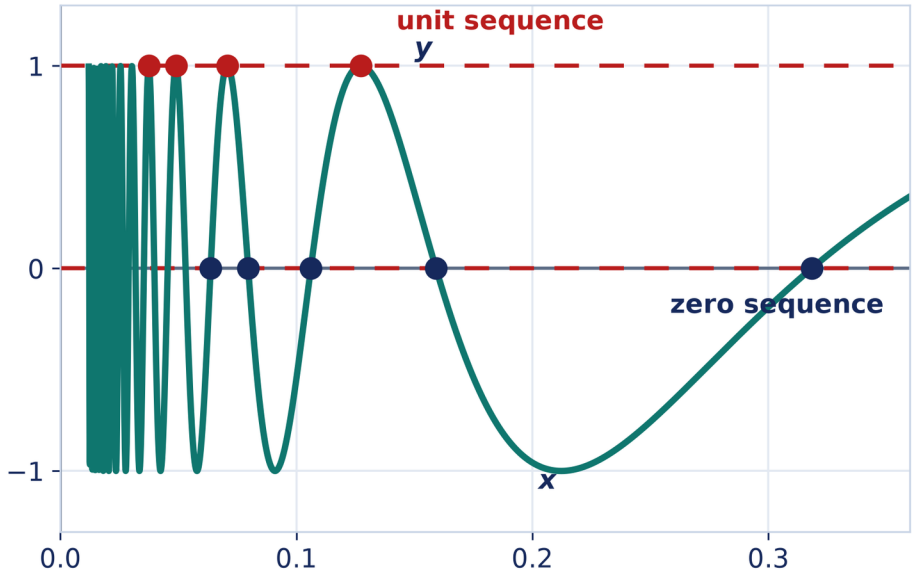
$$x_n = \frac{1}{n\pi}, \qquad y_n = \frac{1}{\pi/2 + 2n\pi}.$$

Both tend to 0 from the right, but

$$\sin\frac{1}{x_n} = \sin(n\pi) = 0, \qquad \sin\frac{1}{y_n} = 1.$$

Thus $\sin(1/x)$ has no limit as $x \to 0$. The oscillation is not a failure of algebra; it is verified nonexistence.

Two marked sequences approach x=0 but keep different heights



READ THE FIGURE

The zero-height points use x_n=1/(n pi); the unit-height points use y_n=1/(pi/2+2n pi). Their inputs all approach 0.

Dense sets can make every approach fail

Define $D(x) = 1$ for rational x and $D(x) = 0$ for irrational x. At any real a, rationals and irrationals are both dense, so there are sequences $q_n \to a$ and $r_n \to a$ with

$$D(q_n) = 1, \qquad D(r_n) = 0.$$

Therefore the Dirichlet function has no limit at any point. This is stronger than having a single jump or removable discontinuity.

A substitution can certify every sequence at once

For $x \neq 0$, put $u = 1/x^2$. Whenever $x_n \to 0$ with $x_n \neq 0$, we have $u_n = 1/x_n^2 \to \infty$, and hence

$$e^{-1/x_n^2} = e^{-u_n} \longrightarrow 0.$$

Heine's criterion now proves the two-sided limit $e^{-1/x^2} \to 0$. Assigning the value 0 at $x = 0$ therefore makes this function continuous there.

KEY RESULT

Existence and nonexistence need different amounts of evidence

One sequence can never prove a full function limit. One pair of sequences can disprove it. For a positive proof, use an estimate valid on a whole punctured neighbourhood or a substitution valid for every admissible sequence.

LESSON 02 / 05

Stolz–Cesàro is a discrete quotient rule

If b_n is strictly increasing and unbounded, and

$$\frac{a_{n+1}-a_n}{b_{n+1}-b_n} \longrightarrow L,$$

then $a_n/b_n \longrightarrow L$. The denominator increments are eventually positive, so the difference quotient controls successive weighted averages. Check the monotonicity and unboundedness before using the rule.

WORKED EXAMPLE

Arithmetic sums and harmonic growth

For $a_n = 1 + 2 + \cdots + n$ and $b_n = n^2$,

$$\frac{a_{n+1}-a_n}{b_{n+1}-b_n} = \frac{n+1}{2n+1} \longrightarrow \frac{1}{2},$$

so $(1 + \cdots + n)/n^2 \to 1/2$.

For $H_n = 1 + 1/2 + \cdots + 1/n$ and $b_n = \ln n$,

$$\frac{H_{n+1}-H_n}{\ln(n+1)-\ln n} = \frac{1/(n+1)}{\ln(1+1/n)} \longrightarrow 1.$$

Thus $H_n/\ln n \to 1$.

A recursive sequence needs convergence before algebra

If $a_{n+1} = f(a_n)$ with continuous f and $a_n \to L$, then passing to the limit gives $L = f(L)$. That equation is the **last** step, not a proof that the sequence converges. A common positive proof is monotone + bounded.

WORKED EXAMPLE

A nested-square-root recurrence

Let $a_1 = 1$ and $a_{n+1} = \sqrt{2 + a_n}$. If $1 \le a_n < 2$, then

$$a_{n+1} = \sqrt{2 + a_n} < 2.$$

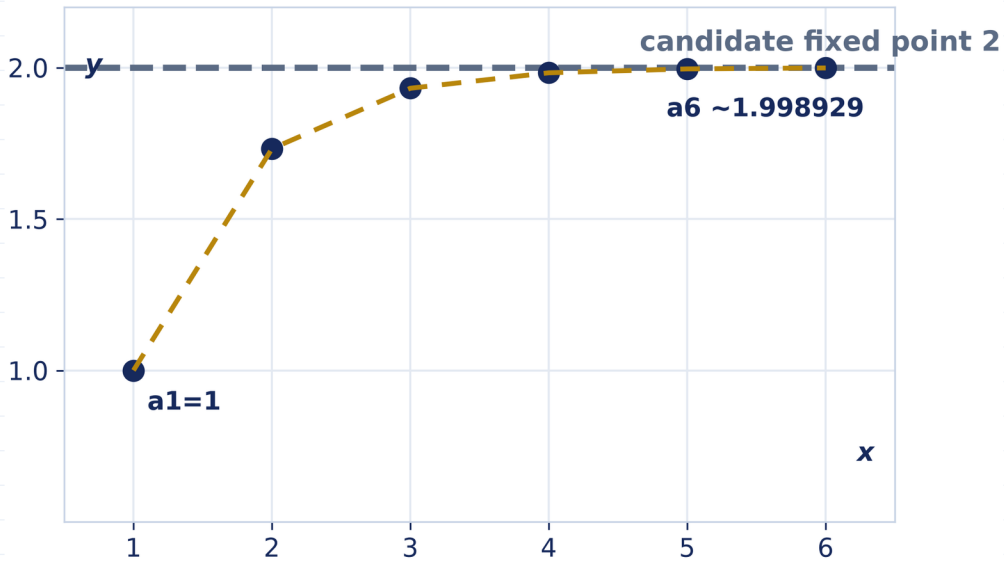
Also $a_{n+1} \ge a_n$ because $2 + a_n \ge a_n^2$ is equivalent to $(2 - a_n)(a_n + 1) \ge 0$. The base case $a_1 = 1$ starts the induction, so the sequence is increasing and bounded above by 2.

Therefore L exists and

$$L = \sqrt{2 + L} \quad \Longrightarrow \quad L^2 - L - 2 = 0.$$

The roots are 2 and -1; positivity selects $L = 2$.

The recurrence increases toward its fixed point 2



READ THE FIGURE
The dots are sequence terms, not a continuous function. Monotone increase and the upper bound prove convergence before the fixed-point equation is used.

WORKED EXAMPLE

Heron's iteration converges downward to sqrt(2)

Let $a_1 = 2$ and $a_{n+1} = (a_n + 2/a_n)/2$. AM–GM gives $a_{n+1} \geq \sqrt{2}$. When $a_n \geq \sqrt{2}$, we also have $2/a_n \leq a_n$, so $a_{n+1} \leq a_n$. Thus the sequence is decreasing and bounded below.

Its limit satisfies $L = (L + 2/L)/2$, so $L^2 = 2$. Positivity gives $L = \sqrt{2}$, not the algebraic alternative $-\sqrt{2}$.

WORKED EXAMPLE

An exponent-tracking recurrence

For $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2a_n}$, induction gives

$$a_n = 2^{1-2^{-n}} \longrightarrow 2.$$

Here an explicit exponent formula replaces the monotone-and-bounded argument.

COMMON TRAP

A fixed-point equation can have extraneous candidates

Writing $L = f(L)$ before proving convergence may produce numbers that the sequence never approaches. Establish monotonicity and a bound, or another convergence argument, then use continuity and discard candidates using sign and range information.

LESSON 03 / 05

Recognize a right-endpoint Riemann sum

For continuous f on $[0,1]$,

$$\frac{1}{n}\sum_{k=1}^{n} f\left(\frac{k}{n}\right) \longrightarrow \int_0^1 f(x)\,dx.$$

The factor $1/n$ is the mesh and k/n are right endpoints. This extension previews the definite-integral theory developed fully in Chapter 21; here it is used as a sequence-limit technique.

WORKED EXAMPLE

Expose the sample point before integrating

$$\sum_{k=1}^{n} \frac{n}{n^2+k^2} = \frac{1}{n}\sum_{k=1}^{n} \frac{1}{1+(k/n)^2} \longrightarrow \int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4}.$$

Similarly,

$$\frac{1}{n}\sum_{k=1}^{n} \frac{1}{1+k/n} \longrightarrow \int_0^1 \frac{dx}{1+x} = \ln 2.$$

For $p>0$, factoring n^p gives

$$\frac{1^p+2^p+\cdots+n^p}{n^{p+1}} = \frac{1}{n}\sum_{k=1}^{n} \left(\frac{k}{n}\right)^p \longrightarrow \frac{1}{p+1}.$$

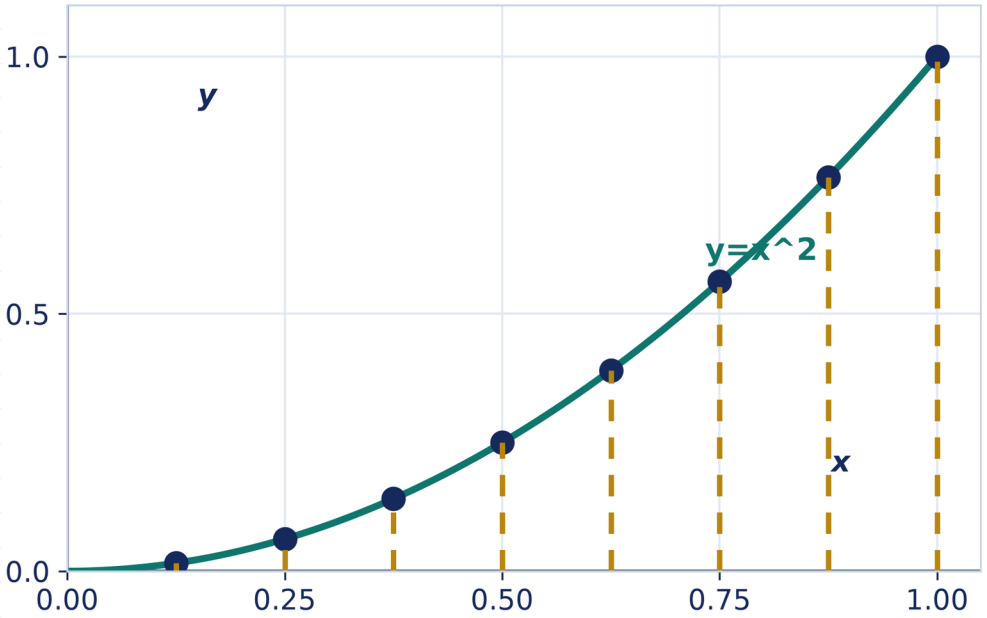
The values $p = 1/2, 1, 2, 3$ give $2/3, 1/2, 1/3, 1/4$ respectively.

KEY RESULT

Keep the mesh, sample point and error order visible

A sum is not a Riemann sum until the factor $1/n$ and the sampled function are explicit. A product is not evaluated by replacing every factor by 1: infinitely many small contributions can add to a nonzero logarithm.

Eight right endpoints approximate the area under y=x^2



READ THE FIGURE

For n=8 the right-endpoint sum is 51/128 = 0.3984375, above the limit 1/3. Increasing n refines the mesh; the samples are not an unweighted average.

Products become sums after logarithms

Let

$$P_n = \prod_{k=1}^{n} \left(1 + \tfrac{k}{n^2}\right).$$

Then $\ln P_n = \sum_{k=1}^{n} \ln(1 + k/n^2)$. For $x \ge 0$,

$$x - \tfrac{x^2}{2} \le \ln(1+x) \le x.$$

Summing the upper bound gives

$$\sum_{k=1}^{n} \tfrac{k}{n^2} = \tfrac{n+1}{2n} \longrightarrow \tfrac{1}{2}.$$

The subtracted error is bounded by

$$\tfrac{1}{2} \sum_{k=1}^{n} \tfrac{k^2}{n^4} = \tfrac{n(n+1)(2n+1)}{12n^4} \longrightarrow 0.$$

Hence $\ln P_n \to 1/2$, and continuity of the exponential gives $P_n \to \sqrt{e}$.

LESSON 04 / 05

Asymptotic equivalence preserves ratios, not differences

Write $u_n \sim v_n$ when $u_n/v_n \to 1$. Stirling's formula states

$$n! \sim \sqrt{2\pi n}\,\left(\frac{n}{e}\right)^n.$$

One proof compares $\ln(n!) = \sum_{k=1}^{n} \ln k$ with an integral and then tracks the correction terms. The formula is especially useful when factorial powers cancel in a ratio.

WORKED EXAMPLE

Central-binomial decay

Applying Stirling to numerator and denominator,

$$(2n)! \sim \sqrt{4\pi n}\,\left(\frac{2n}{e}\right)^{2n}, \qquad (n!)^2 \sim 2\pi n\left(\frac{n}{e}\right)^{2n}.$$

Therefore

$$\frac{(2n)!}{4^n (n!)^2} \sim \frac{1}{\sqrt{\pi n}}.$$

In particular the ratio tends to 0, while n times its square tends to $1/\pi$. Merely writing every factorial as "large" would lose this exact decay rate.

Harmonic sums separate a logarithm from a constant

The Euler–Mascheroni constant is defined by

$$H_n - \ln n \longrightarrow \gamma \approx 0.5772156649.$$

Consequently $H_n/\ln n \to 1$, but the difference $H_n - \ln n$ does **not** tend to 0. This illustrates why asymptotic equivalence cannot automatically be subtracted.

WORKED EXAMPLE

Cancel gamma without assuming it is zero

Since $\ln(2n) = \ln 2 + \ln n$,

$$H_{2n} - H_n = (H_{2n} - \ln(2n)) - (H_n - \ln n) + \ln 2 \longrightarrow \gamma - \gamma + \ln 2 = \ln 2.$$

Thus $H_{2n} - H_n - \ln 2 \to 0$, while $H_{2n} - H_n$ itself tends to $\ln 2$.

A second-order term determines the convergence rate

The notation $R_n = O(n^{-2})$ means $|R_n| \le Cn^{-2}$ for some constant C and all large n. Starting with

$$\ln(1+x) = x - \tfrac{x^2}{2} + O(x^3) \qquad (x \to 0),$$

put $x = 1/n$ and multiply by n:

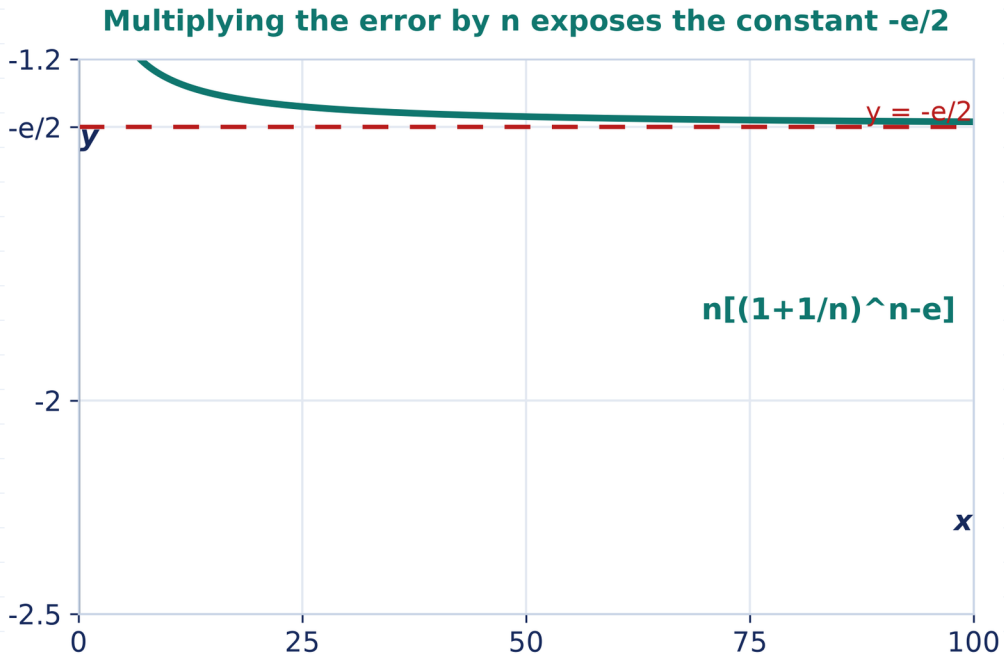
$$n\ln\left(1+\tfrac{1}{n}\right) = 1 - \tfrac{1}{2n} + O(n^{-2}).$$

Exponentiating and expanding $e^u = 1 + u + O(u^2)$ gives

$$\left(1+\tfrac{1}{n}\right)^n = e\left(1 - \tfrac{1}{2n} + O(n^{-2})\right).$$

Therefore

$$n\left[\left(1+\tfrac{1}{n}\right)^n - e\right] \longrightarrow -\tfrac{e}{2}.$$



READ THE FIGURE
The unscaled error tends to 0, but the scaled error tends to -e/2. A graph of the original sequence alone would hide this first-order rate.

COMMON TRAP
The leading limit and the rate answer different questions
The statements $(1+1/n)^n \to e$ and $n[(1+1/n)^n - e] \to -e/2$ are compatible. The second contains more information. To obtain it, expand one order beyond the first constant term and keep the Big-O remainder under control.

LESSON 05 / 05

Build a function whose endpoint signs are forced to oppose

Suppose f is continuous on $[0,1]$ and $f(0)=f(1)$. Define

$$g(x)=f(x)-f\left(x+\tfrac{1}{2}\right), \qquad 0\le x\le \tfrac{1}{2}.$$

Then

$$g\left(\tfrac{1}{2}\right)=f\left(\tfrac{1}{2}\right)-f(1)=f\left(\tfrac{1}{2}\right)-f(0)=-g(0).$$

If $g(0)=0$, take $c=0$. Otherwise IVT gives $c\in(0,1/2)$ with $g(c)=0$. Hence

$$f(c)=f\left(c+\tfrac{1}{2}\right).$$

WORKED EXAMPLE

An exact half-shift check

For $f(x)=(x-1/2)^2$, we have $f(0)=f(1)=1/4$ and

$$g(x)=\left(x-\tfrac{1}{2}\right)^2-x^2=\tfrac{1}{4}-x.$$

Thus $c=1/4$, and indeed $f(1/4)=f(3/4)=1/16$. The same antipodal-difference idea applied to continuous temperature T on a circle gives at least one pair of antipodal points with equal temperature.

Continuity extends rational information to all reals

An additive function satisfies $f(x+y)=f(x)+f(y)$. Induction and scaling give

$$f(qx)=qf(x) \qquad (q\in\mathbb{Q}).$$

If f is continuous and $a=f(1)$, then for any real x, choose rationals $q_k\to x$. Continuity gives

$$f(x)=\lim_{k\to\infty} f(q_k)=\lim_{k\to\infty} aq_k=ax.$$

Without continuity, additivity alone permits pathological solutions.

For the Jensen midpoint relation

$$f\left(\frac{x+y}{2}\right)=\frac{f(x)+f(y)}{2},$$

subtract $b=f(0)$. The function $g=f-b$ has the same midpoint property and $g(0)=0$. It follows that g is additive, so continuity at one point gives $g(x)=ax$ and therefore $f(x)=ax+b$.

WORKED EXAMPLE

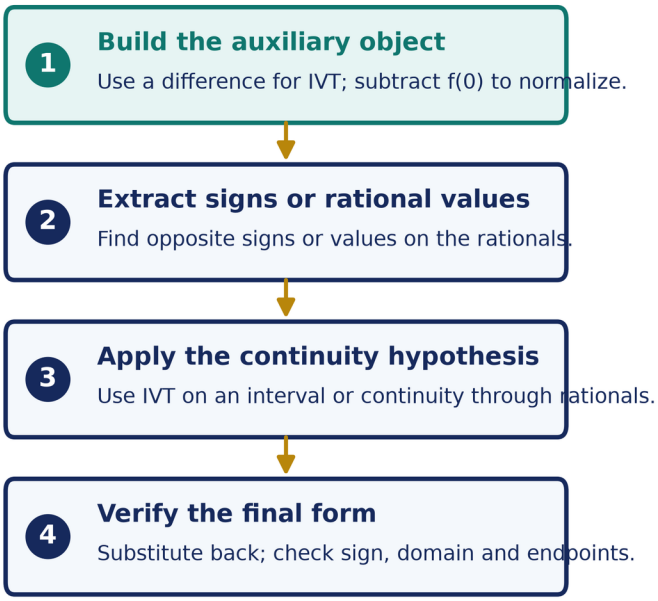
The continuous exponential functional equation

Suppose f is continuous, $f(x+y) = f(x)f(y)$, and $f(1) = a > 0$. First, $f(0) = 1$; the alternative $f(0) = 0$ would force $f \equiv 0$, contradicting $f(1) > 0$. Also $f(x) = f(x/2)^2 \geq 0$, and a zero value anywhere would force $f(1) = 0$, so f is strictly positive.

The functional equation gives $f(q) = a^q$ for every rational q. Rational density and continuity then extend this to

$$f(x) = a^x \qquad (x \in \mathbb{R}).$$

Continuity turns local or discrete information into a global conclusion



READ THE FIGURE

The algebra produces candidates; continuity is what makes the limiting or density argument legitimate.

KEY RESULT

Do not overstate antipodal conclusions

The argument guarantees ***at least one*** antipodal pair with equal values, not that every attained value occurs at antipodal pairs. Likewise, IVT guarantees a point with a required value, while a separate argument is needed for uniqueness or a formula for that point.